(A)
Environment description

I selected FrozenLake with the 8x8 map.

The environment's objective to to mobe from the starting tile to the goal time while avoiding holes, where the frozen tiles are safe to walk on. Reaching the goal rewards 1 and all other transitions reward 0. Going into a hole ends the run with no reward.

There are 64 possible states in the 8x8 grid. The observation space is Discrete(64).

There are 4 possible actions: left, right, up, and down.

If the frozen lake is slippery, the direction that it transitions one space in is random. If the lake is not slippery, it transitions to the next space in the intended direction.

The reward structure is earning 1 point by reaching the goal, otherwise earning 0 points.

The session will be terminated when the goal is reached or when the agent moves onto a hole space.

Settings

- Environment: FrozenLake-v1
- Map size: 8x8
- Number of states: 64
- Number of actions:4
- Slippery transitions: disabled
- State representation: discrete integer
- Q-table dimensions: 64x4

(C)
Baseline Training:
Learning rate: 0.1
Discount factor: 0.99
Initial exploration rate: 1.0
Minimum exploration rate: 0.01
Exploration decay rate: -.9999
Number of training episodes: 50000
Maximum steps per episode: 100
Slippery transitions: disabled
Schedule: exponantial decay
The graph shows clear improvement in performance over training, success rate from 0.106 to 0.985

Output:

Episode: 5000, Success rate (last 5k): 0.106, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.386, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.622, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.788, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.873, Epsilon: 0.082

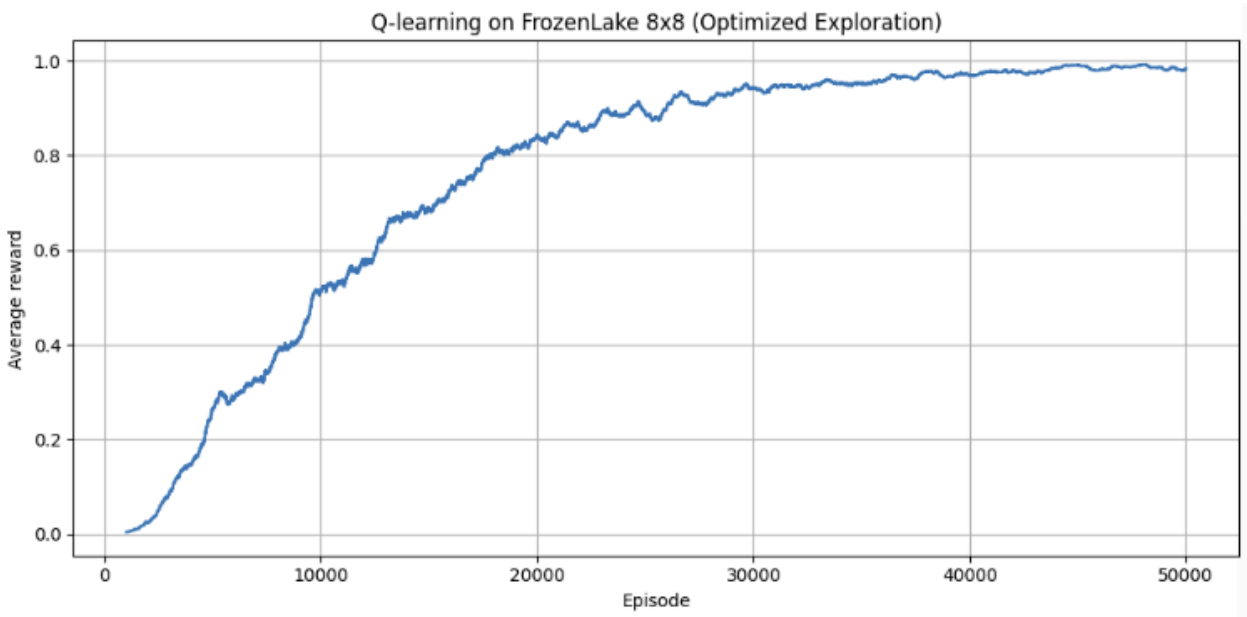
Episode: 30000, Success rate (last 5k): 0.923, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.950, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.968, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.982, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.985, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 50 / 64

(D)

Learning-rate experiment:
The 0.2 learning rate performed better in early learning, with 0.5 and 0.9 catching up shortly after. The 0.5 learning rate finished with the highest success rate by a small margin. Overall, 0.5 is the best because it had the highest success rate in the end, but they all performed similarly.

Learning rate = 0.2

Episode: 5000, Success rate (last 5k): 0.077, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.256, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.490, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.673, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.813, Epsilon: 0.082

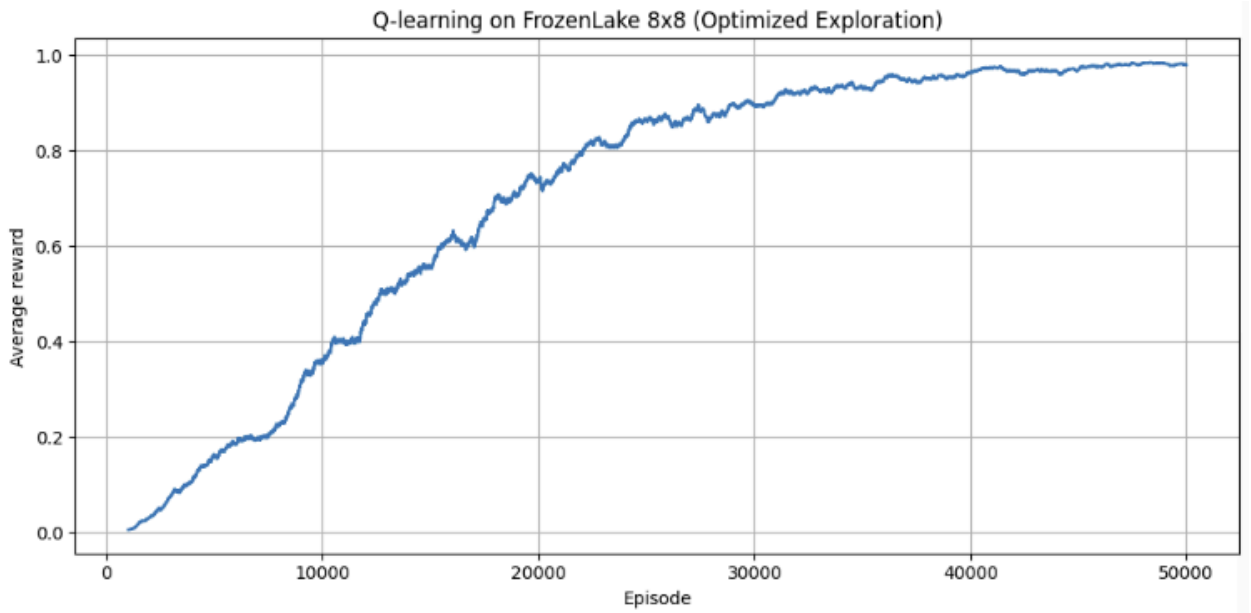
Episode: 30000, Success rate (last 5k): 0.880, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.926, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.954, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.970, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.981, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 50 / 64

Learning rate = 0.5

Episode: 5000, Success rate (last 5k): 0.054, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.245, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.486, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.679, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.812, Epsilon: 0.082

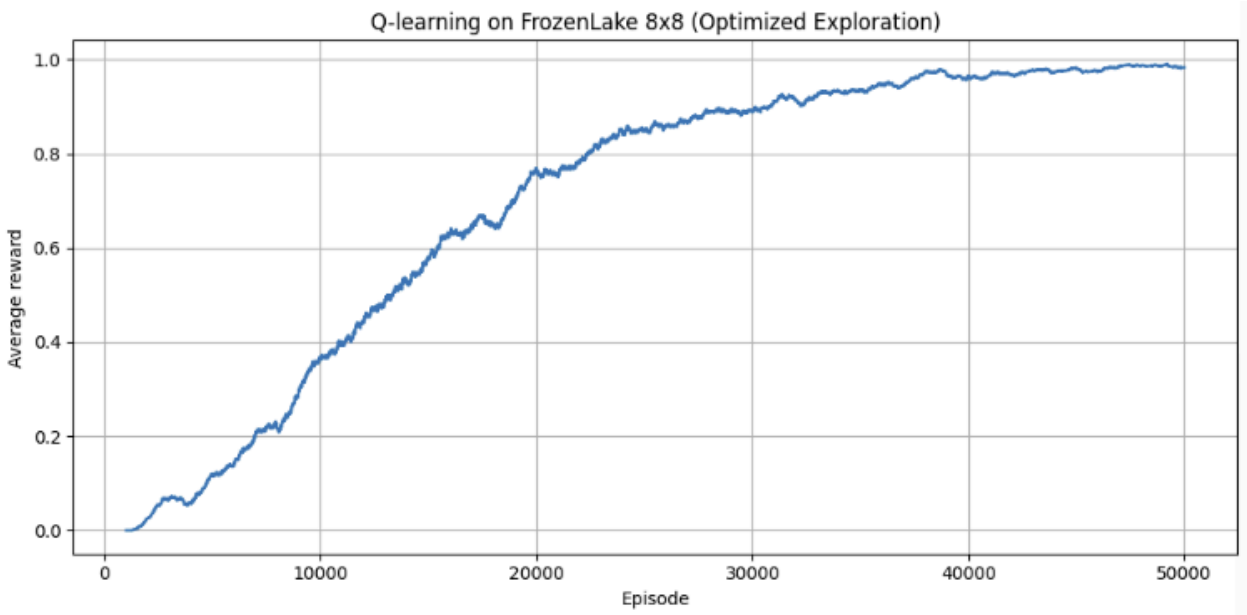
Episode: 30000, Success rate (last 5k): 0.879, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.923, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.959, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.972, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.984, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 49 / 64

Learning rate = 0.9

Episode: 5000, Success rate (last 5k): 0.047, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.254, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.497, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.676, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.805, Epsilon: 0.082

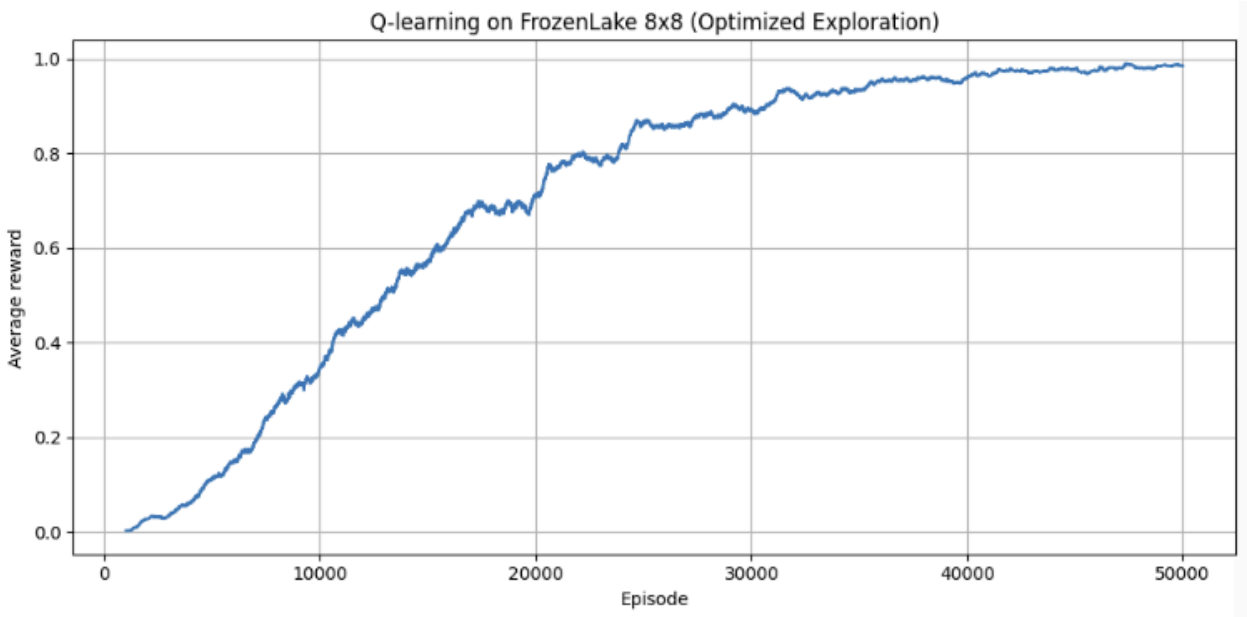
Episode: 30000, Success rate (last 5k): 0.878, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.926, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.957, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.974, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.982, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 50 / 64

(E)
Discount-factor experiment

0.75 did best in early learning, with 0.50 and 0.90 catching up shortly after. 0.50 finished with the highest success rate by a narrow margin. Overall, 0.50 is the best because it ended with the highest success rate, but they are all performed similarly.

Discount-factor = 0.50

Episode: 5000, Success rate (last 5k): 0.066, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.279, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.502, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.682, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.804, Epsilon: 0.082

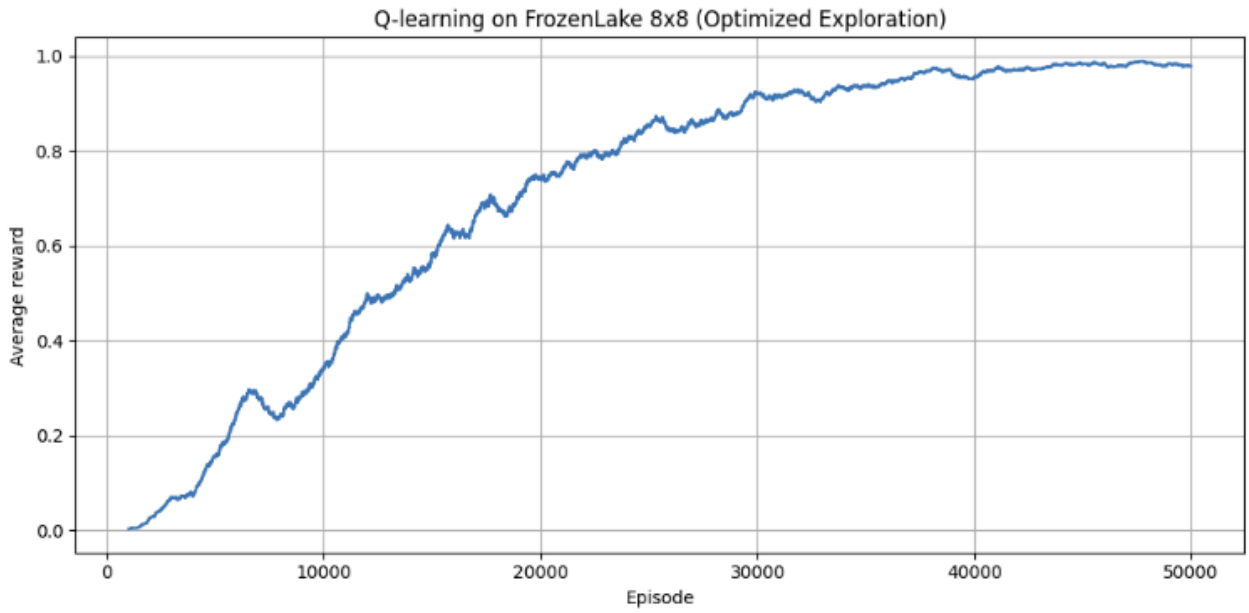
Episode: 30000, Success rate (last 5k): 0.874, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.925, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.958, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.977, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.983, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 50 / 64

Discount Factor = 0.75

Episode: 5000, Success rate (last 5k): 0.101, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.375, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.498, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.681, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.806, Epsilon: 0.082

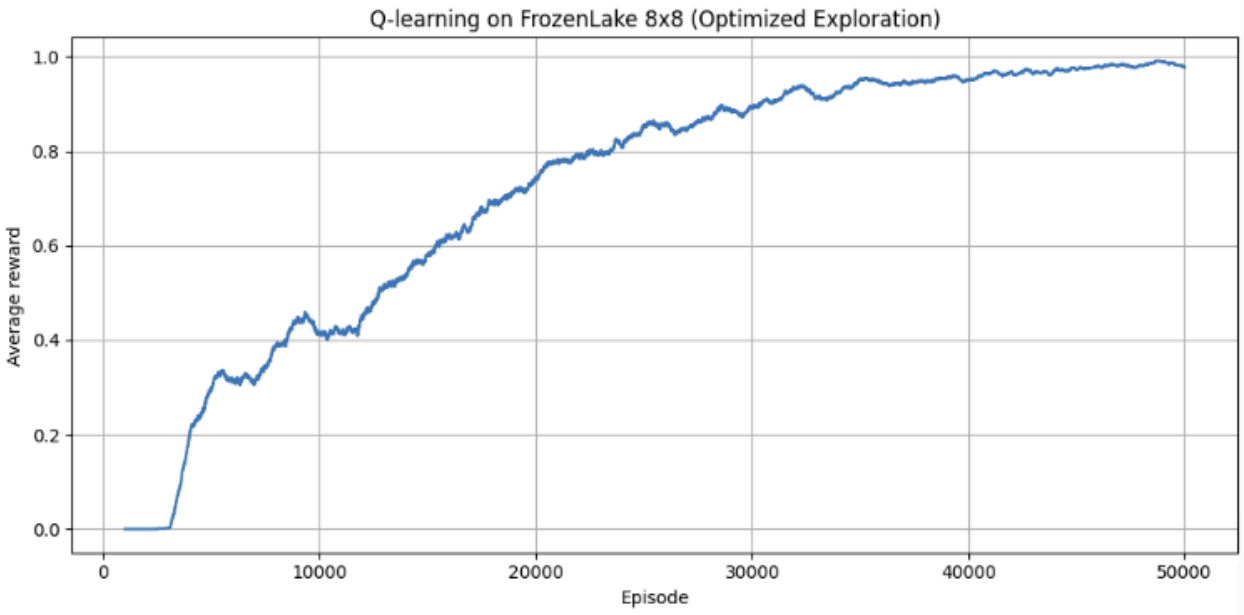
Episode: 30000, Success rate (last 5k): 0.871, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.925, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.949, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.968, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.982, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 47 / 64

**Discount Factor = 0.90

Episode: 5000, Success rate (last 5k): 0.094, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.321, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.498, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.677, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.801, Epsilon: 0.082

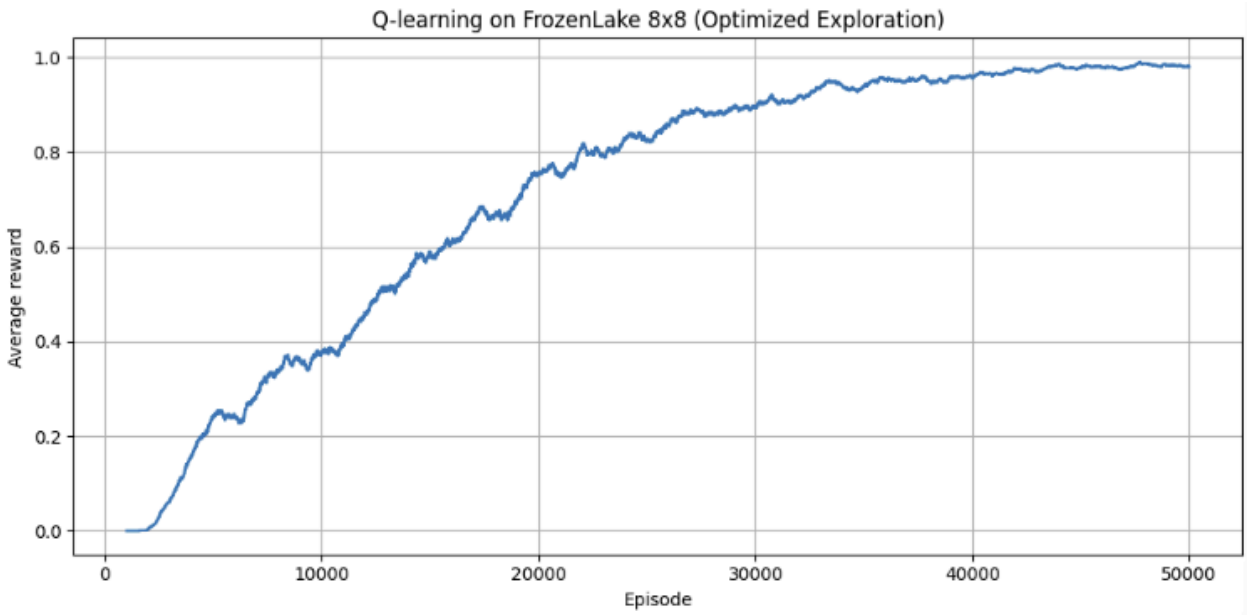
Episode: 30000, Success rate (last 5k): 0.883, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.927, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.953, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.974, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.980, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 48 / 64

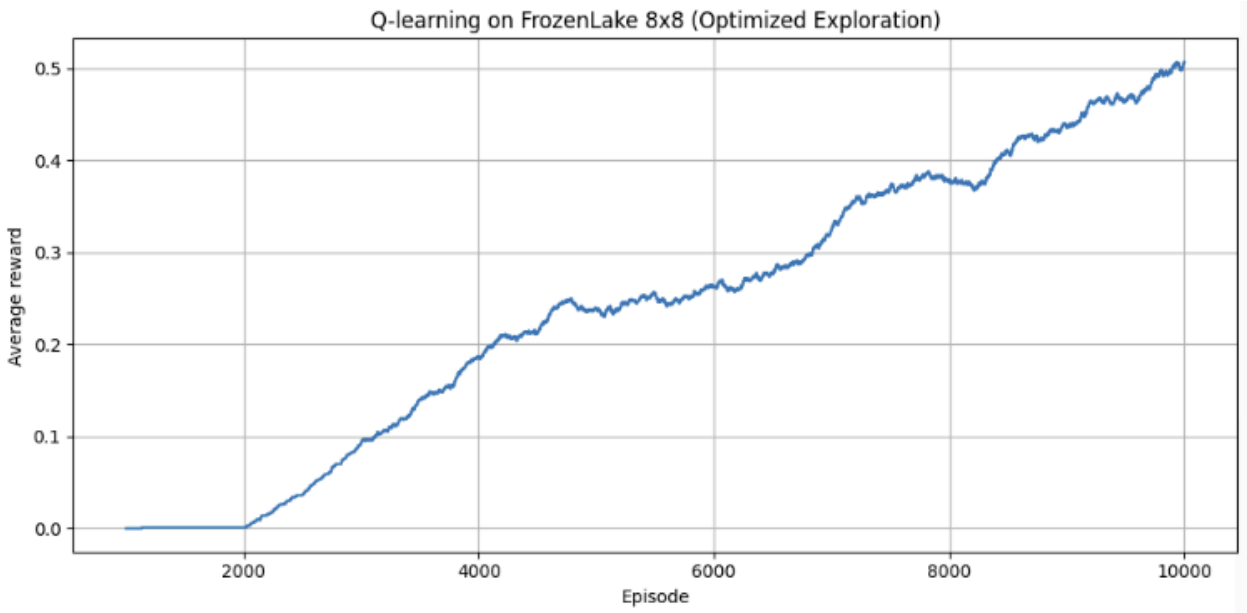
(F)
Episodes Experiment
They all perform similarly in early learning, with 90000 doing the best by a small measure. They all had a substantial increase in learning, but 10000 was unable to reach a success rate over 0.382 due to lack of episodes to work with. 50000 and 90000 reach a high success rate, but

seem to planteau at this high success rate. Due to this dual plateau at around 98.6%, 90000 is not needed, and 50000 is sufficient and best for this model.

Episodes = 10000

Episode: 5000, Success rate (last 5k): 0.104, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.382, Epsilon: 0.368




Learned Q-table (non-zero entries indicate path found):

States with learned values: 48 / 64

Episodes = 50000

Episode: 5000, Success rate (last 5k): 0.094, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.370, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.562, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.682, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.801, Epsilon: 0.082

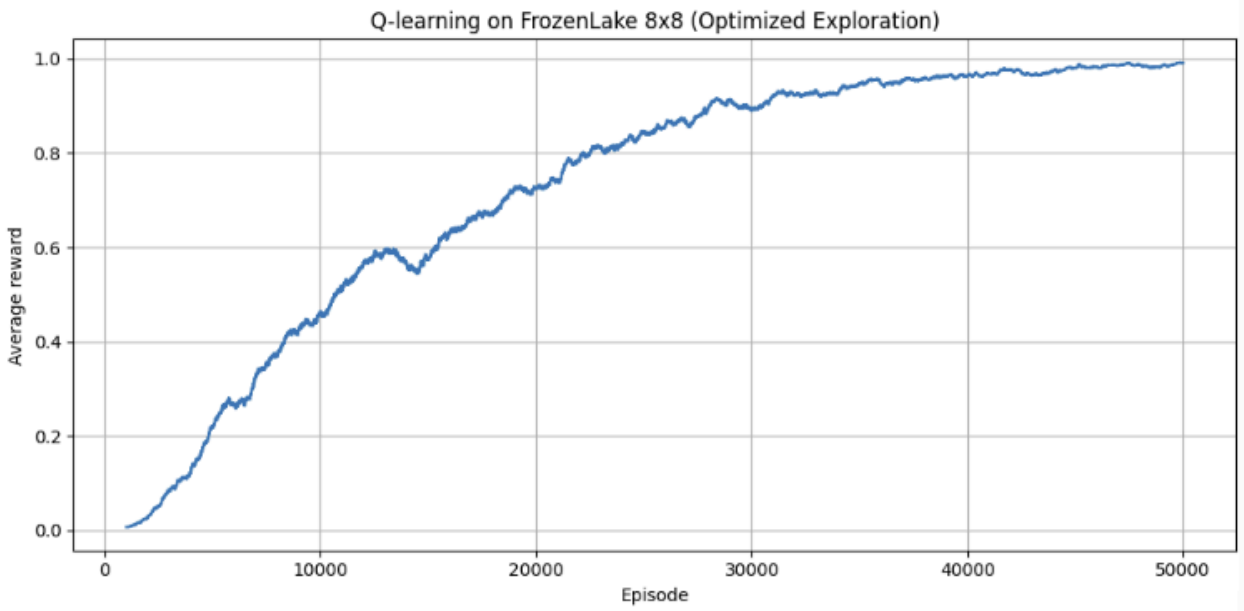
Episode: 30000, Success rate (last 5k): 0.884, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.930, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.957, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.973, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.986, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 48 / 64

Episodes = 90000

Episode: 5000, Success rate (last 5k): 0.121, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.385, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.628, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.776, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.866, Epsilon: 0.082

Episode: 30000, Success rate (last 5k): 0.916, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.951, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.972, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.983, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.988, Epsilon: 0.010

Episode: 55000, Success rate (last 5k): 0.989, Epsilon: 0.010

Episode: 60000, Success rate (last 5k): 0.989, Epsilon: 0.010

Episode: 65000, Success rate (last 5k): 0.988, Epsilon: 0.010

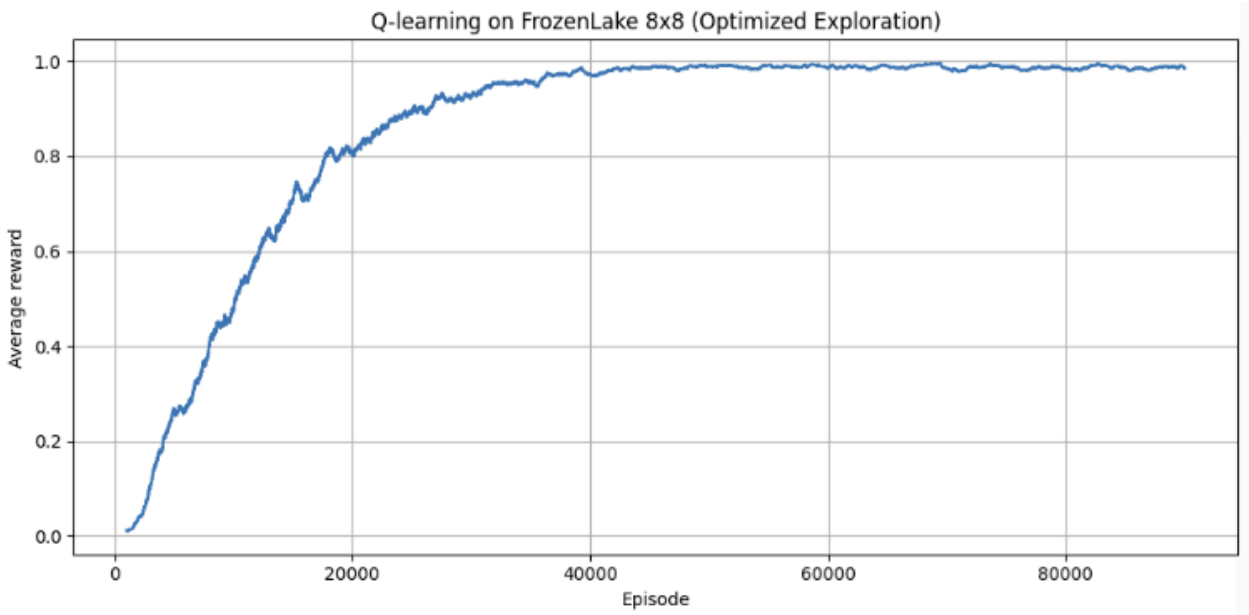
Episode: 70000, Success rate (last 5k): 0.990, Epsilon: 0.010

Episode: 75000, Success rate (last 5k): 0.987, Epsilon: 0.010

Episode: 80000, Success rate (last 5k): 0.986, Epsilon: 0.010

Episode: 85000, Success rate (last 5k): 0.987, Epsilon: 0.010

Episode: 90000, Success rate (last 5k): 0.986, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):

States with learned values: 49 / 64

(G)

Best Agent
Parameters

The following are the parameters that I determined to be the best, based on the experiments performed on learning rate, discount factor, and number of episodes.

Learning rate: 0.5
Discount factor: 0.50
Initial exploration rate: 1.0
Minimum exploration rate: 0.01
Exploration decay rate: -.9999
Number of training episodes: 50000
Maximum steps per episode: 100
Slippery transitions: disabled
Schedule: exponantial decay
Evaluation

This model proved to be successful, reaching 0.98 success rate by the end, and learning well early.

Results

Episode: 5000, Success rate (last 5k): 0.050, Epsilon: 0.607

Episode: 10000, Success rate (last 5k): 0.241, Epsilon: 0.368

Episode: 15000, Success rate (last 5k): 0.482, Epsilon: 0.223

Episode: 20000, Success rate (last 5k): 0.669, Epsilon: 0.135

Episode: 25000, Success rate (last 5k): 0.794, Epsilon: 0.082

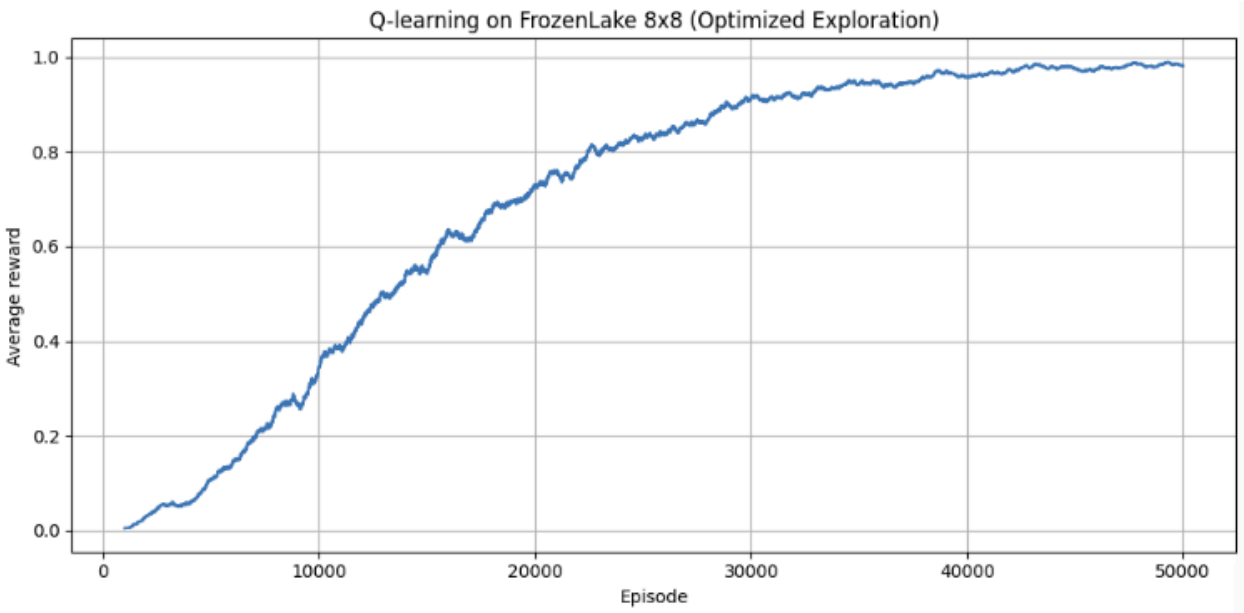
Episode: 30000, Success rate (last 5k): 0.875, Epsilon: 0.050

Episode: 35000, Success rate (last 5k): 0.927, Epsilon: 0.030

Episode: 40000, Success rate (last 5k): 0.953, Epsilon: 0.018

Episode: 45000, Success rate (last 5k): 0.974, Epsilon: 0.011

Episode: 50000, Success rate (last 5k): 0.980, Epsilon: 0.010



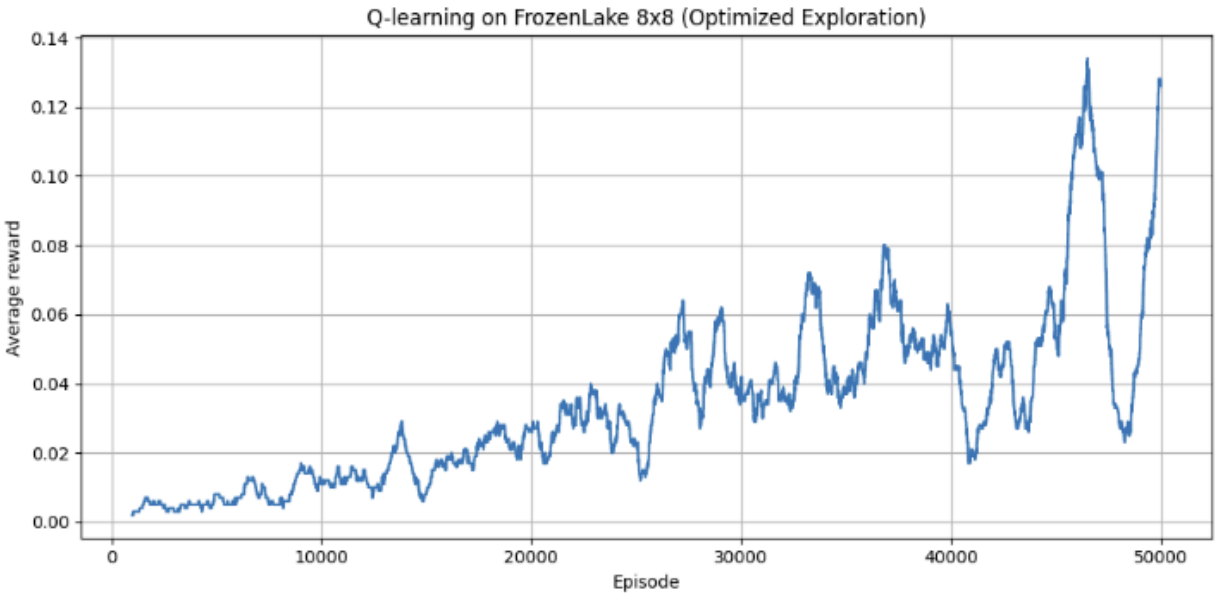

Learned Q-table (non-zero entries indicate path found):

States with learned values: 51 / 64

(H)

Stochastic environment
Turning slippery tiles on makes it so that there is a 1/3 chance that the agent does not move to its intended tile, making it more likely to fail by going into a hole. I kept all parameters the same as the "best" version of the original model. The stochastity caused the rate of success to increase more slowly, reaching 0.070, in comparison to the 0.980 of this model minus its stochastity. The Q-learning graph is also a lot more jagged due to the random actions leading to failure.

Results:
Episode: 5000, Success rate (last 5k): 0.005, Epsilon: 0.607
Episode: 10000, Success rate (last 5k): 0.009, Epsilon: 0.368
Episode: 15000, Success rate (last 5k): 0.014, Epsilon: 0.223
Episode: 20000, Success rate (last 5k): 0.022, Epsilon: 0.135
Episode: 25000, Success rate (last 5k): 0.027, Epsilon: 0.082
Episode: 30000, Success rate (last 5k): 0.044, Epsilon: 0.050
Episode: 35000, Success rate (last 5k): 0.043, Epsilon: 0.030
Episode: 40000, Success rate (last 5k): 0.056, Epsilon: 0.018
Episode: 45000, Success rate (last 5k): 0.040, Epsilon: 0.011
Episode: 50000, Success rate (last 5k): 0.083, Epsilon: 0.010




Learned Q-table (non-zero entries indicate path found):
States with learned values: 53 / 64